

Proposal
Of
**Enhancing Product Quality Through Vibration
Signals in Turning Operations**

A project submitted
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1. Introduction

In modern manufacturing, ensuring high product quality is critical, especially in precision machining processes such as CNC turning. One of the primary indicators of product quality in such operations is **surface roughness**, which reflects the condition of the cutting tool, the stability of the machine, and the consistency of the machining parameters. Traditionally, surface roughness is evaluated through post-process inspection using contact-based or optical measurement systems. However, these methods are time-consuming, limited in scope, and reactive rather than proactive.

To overcome these challenges, the use of vibration signals as a real-time indicator of machining quality has gained significant attention. During a turning operation, vibrations are inherently generated due to interactions between the cutting tool and the workpiece. These vibrations carry rich information about the cutting condition, tool wear, material deformation, and surface finish. By analysing these signals using advanced signal processing and modelling techniques, it becomes possible to predict surface quality in real time, enabling proactive decision-making and adaptive control in innovative manufacturing systems.

In this project, we focus on enhancing product quality in turning operations by leveraging vibration signal analysis. Acceleration signals from three axes (X, Y, Z) are captured using sensors mounted on the machine tool. These signals undergo preprocessing and are segmented into time windows. We then extract a variety of features from time-domain, frequency-domain (using FFT), and time-frequency domain (using short-time fourier transform) techniques.

To establish a quantitative relationship between these vibration-based features and the actual surface roughness (R_a), we employ **multilinear regression modeling**. This technique allows us to evaluate how combinations of features contribute to the prediction of surface roughness by maximizing the R^2 score. The regression model helps us identify the most significant features and their interactions, offering a statistically sound method for feature selection and prediction.

In addition to regression, we also implement a **fuzzy logic-based prediction model**. Fuzzy logic is particularly useful in this context because it handles uncertainty and imprecision inherent in machining processes. Unlike traditional models that rely on crisp thresholds and hard rules, fuzzy logic allows for soft classification of feature values through membership functions and linguistic rules. This enables more human-like reasoning and can provide better generalization, especially when the data has variability or noise.

By integrating both statistical modeling (multilinear regression) and intelligent systems (fuzzy logic), our approach aims to develop a robust predictive framework that not only correlates vibration signals with product quality but also supports real-time quality monitoring in CNC turning operations.

2. Objectives

The primary goal of this project is to enhance product quality in CNC turning operations through the analysis of vibration signals. To achieve this, the project is structured around the following key objectives:

1. To capture real-time vibration signals during CNC turning operations

- Acquire tri-axial acceleration data (X, Y, Z) using sensors mounted on the machine setup.
- Ensure signal collection under varied machining conditions to represent diverse surface roughness outcomes.

2. To preprocess and segment the vibration signals for feature extraction

- Filter the time series data to eliminate noise and focus on steady-state cutting (typically between 1.0s to 4.0s).
- Apply windowing techniques for consistent feature computation across parts.

3. To extract significant features across different domains

- Time-domain: RMS, Peak, Crest Factor, Skewness, Kurtosis.
- Frequency-domain: Power Spectral Density, Dominant Frequency.
- Time-frequency domain: Spectral Entropy, Wavelet Kurtosis, etc.

4. To select the most relevant features using statistical analysis

- Compute R^2 correlation scores for each feature against the surface roughness (Y_{true}).
- Select the top 12 features and evaluate all possible 4-feature combinations for model input.

5. To develop a multilinear regression model for surface roughness prediction

- Train the regression model using selected features.
- Evaluate model performance using accuracy metrics.

6. To implement a fuzzy logic-based model for predicting surface roughness

- Design membership functions and rules based on selected features.
- Compare fuzzy logic predictions with actual surface roughness values.
- Evaluate the accuracy of the fuzzy model under varying input conditions.

3. Assumptions and Constraints

Assumptions

- The CNC machine operates under stable temperature and mechanical conditions.
- Workpiece and tool material properties remain consistent throughout the trials.
- Accelerometers are properly calibrated and accurately capture vibration data.
- Surface roughness measurements (Y_{true}) are accurate and consistent.
- Fuzzy logic is suitable for modeling uncertainty in vibration-based quality prediction.

Constraints

- **Experimental Analysis**
 - ✓ The experimental setup is limited to a controlled machining environment, which may not represent real-world industrial variability.
 - ✓ External influences like temperature fluctuations or tool wear progression during longer production cycles are not evaluated.
- **Dataset Information**
 - ✓ The dataset used in the project is utilized vibration data from 27 parts for training and 16 parts for testing consisting of a limited number of parts and parameters.
 - ✓ The real solution is by extracting the features and finding the most relevant features to perform the model.
 - ✓ This restricts the model's ability to generalize across a broader range of machining conditions or part geometries.
- **Data Analysis Technology and Prediction Modeling**
 - ✓ Through statistical, data, and regression analysis, we pinpointed the four features which used to perform model. Following the analysis of these top four features, we made predictions for dependent variables(Y) and evaluated in the model accuracy.
- **Regression Model Limitations**
 - ✓ A multilinear regression model was developed using the top four vibration features selected based on R^2 scores.
 - ✓ The challenge lies in accurately predicting the accuracy based on these variables.
 - ✓ This assumes a linear relationship between inputs and surface roughness, which may oversimplify the true machining dynamics.
- **Creating Rule Bank Using Data Sets**
 - ✓ The fuzzy logic rule base is manually crafted using selected feature combinations.

- ✓ The rule generation process is static and lacks adaptive tuning based on incoming data, which limits scalability.

- **Calculation of Membership Functions**
 - ✓ Membership functions are generated using manually defined input ranges (min, max, average), and are not self-optimized.
 - ✓ This task is accomplished in the membership function called calculateMU, where it takes the input of min, max and avg. values from row iteration values of corresponding x variable values and internally calculates the membership values and returns the mu_values whether it consists of S and M, or M and H.

- **Calculation of diameter prediction[y_pred] and Accuracy**
 - ✓ The goal is to calculate the y_pred and final accuracies of the y_pred and y.
 - ✓ With the help of the function named calculateAccuracy we are using the predefined functions calculate MU and rules which calculates the MU values and gives us the rules for each iteration.
 - ✓ Basing on these values we calculate the y_pred values. By using these y_pred values we further calculate the accuracy using the fuzzy logic formula.

4. Methodology

The methodology followed in this project involves a systematic approach to predict surface roughness using vibration signals collected from CNC turning operations.

4.1 Data Acquisition using CNC Lathe Machine

- The experimental data for this study was collected using a CNC lathe machine, where cylindrical workpieces were machined under controlled conditions.
- A total of 43 parts were produced by varying machining parameters such as spindle speed, feed rate, and depth of cut to create a diverse range of surface finish qualities. Of these, 27 parts were used for model training, and the remaining 16 parts were reserved for testing.
- During each machining operation, tri-axial vibration signals (X, Y, Z directions) were captured in real time using accelerometers mounted on the machine tool. These vibration signals served as the raw input data for further signal processing and analysis.

4.2 Surface Roughness Measurement (Y_True)

- Post-machining, the surface roughness (Ra) of each part was measured using a surface roughness machine.
- For accuracy, three measurements were taken from different locations on each part, and the average value was recorded as the ground truth (Y_True) for that sample.

4.3 Feature Extraction from Vibration Signals

The raw vibration signals collected were pre-processed and features were extracted across three signal analysis domains:

4.3.1 Time Domain

Time-domain analysis involves analyzing raw signals over time — that is, observing how a signal changes with time without converting it to frequency or other domains.

- **Root Mean Square (RMS):** It is a feature which provides a measure of the energy content in the signal by calculating the square root of the average of the squared signal values. RMS reflects the overall power of the signal and is widely used to assess general vibration severity. The mathematical representation of this is as follows

$$RMS = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N x_i^2}$$

- **Peak Value:** The peak value represents the maximum absolute amplitude in the signal and is useful for identifying sudden impacts or transient events. It is sensitive to anomalies and often indicates the presence of mechanical shocks, such as cracks or misalignments.

$$Peak = \max(|x(t)|)$$

- **Crest Factor:** Crest factor is derived by dividing the peak value by the RMS. It quantifies the ratio of the highest peak to the overall energy level of the signal. A high crest factor typically suggests the presence of impulses or sharp transients, which are characteristic of faults such as bearing defects or looseness.

$$Crest\ Factor = \frac{peak}{RMS}$$

- **Kurtosis:** Kurtosis measures the sharpness or peakedness of the signal distribution by evaluating the fourth standardized moment. It captures how often extreme values occur relative to a normal distribution. Signals with high kurtosis usually contain infrequent but extreme peaks, indicating potential faults or defects in the machinery.

$$Kurtosis = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^4}{\left(\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2\right)^2}$$

4.3.2 Frequency Domain (FFT)

The frequency domain(FFT) analyzes how different vibration frequencies contribute to the signal. It helps in identifying periodic vibration patterns that might indicate chatter or structural resonances. By analyzing the signal in the frequency domain, we can isolate key frequencies related to defective machining conditions.

- **Power Spectral Density (PSD)** quantifies how the signal's power is distributed across frequency. It highlights energy concentration and helps detect abnormal vibrations or imbalances.

$$PSD(f) = \frac{|x(f)|}{N}$$

- **Dominant Frequency** pinpoints the frequency with the highest amplitude, often corresponding to a mechanical fault or system resonance.

$$f_d = \arg \max PSD(f)$$

- **Total Energy** in the frequency domain captures the overall energy content, offering a comprehensive measure of vibration intensity.

$$E_{freq} = \sum_{f=0}^F |x(f)|^2$$

- **Spectral Entropy** reflects the complexity or disorder in the frequency components. A higher entropy value indicates more randomness, which may be associated with unpredictable or degraded machine conditions.

$$H = - \sum_{f=0}^F P(f) \log p(f)$$

4.3.3 Time-Frequency Domain (STFT)

The Short-Time Fourier Transform (STFT) is a powerful signal processing technique that extends the capabilities of FFT by analyzing how frequency content evolves. STFT provides a way to track frequency variations over time, unlike FFT, which provides only a static frequency representation.

- **Spectral Entropy** in STFT measures the randomness or complexity of frequency components over time, helping identify erratic or unstable behaviors in machine operation.

$$H(t) = - \sum_{f=0}^F P(f, t) \log P(f, t)$$

- **Spectral Centroid** indicates the average frequency or the "center of mass" of the spectrum in each window. It shifts higher or lower depending on the dominant frequency content in time.

$$C(t) = \frac{\sum_f f \cdot |X(f, t)|}{\sum_f |X(f, t)|}$$

- **Spectral Bandwidth** describes the spread of the frequencies at each point in time, revealing how wide or narrow the energy is distributed. Broader bandwidths often signal complex or fault-induced vibrations.

$$B(t) = \sqrt{\frac{\sum_f (f - C(t))^2 \cdot |X(f, t)|}{\sum_f |X(f, t)|}}$$

- **Spectral Kurtosis** measures the peakedness of the frequency distribution over time, making it sensitive to transient impulses or extreme vibrations that are important in fault diagnosis.

$$K(t) = \frac{\frac{1}{N} \sum_f (X(f, t) - \mu_f)^4}{\left(\frac{1}{N} \sum_f (X(f, t) - \mu_f)^2 \right)^2}$$

4.4 Feature Selection using R² Analysis

- The extracted features were analyzed using regression analysis in Minitab software to compute the R² (coefficient of determination) values between each feature and surface roughness (Y_True).
- Based on this analysis, the top 12 features showing the highest correlation were selected for further modeling.

4.5 Feature Combination and Selection

- All possible 4-feature combinations from the top 12 were generated using combinatorial analysis. Each combination was evaluated by building a regression model and testing its prediction accuracy.
- The best 4-feature combination (i.e., the one with the highest accuracy) was selected to proceed with model development.

Stage 1: Multilinear Regression Modeling (MLR)

4.6 Regression Model Development

- A Multilinear Regression (MLR) model was developed using the selected 4-feature combination as input variables (X_1 to X_4) and the surface roughness (Y_True) as the output variable.
- The model was trained using the training dataset (27 parts) and tested on unseen test data (16 parts).

4.7 Prediction and Accuracy Calculation

- The model predicted surface roughness values (Y_Pred) for test samples.
- Accuracy was evaluated by comparing Y_Pred with Y_True using standard metrics such as R^2 score and a custom-defined fuzzy accuracy formula.

$$Accuracy = 100 - \left(\left| \frac{Y_true - Y_pred}{Y_true} \right| \right)$$

Stage 2: Fuzzy Logic Model Development

4.8 Fuzzification: Membership Function Definition

- In this stage, each of the selected four input features was mapped to three linguistic levels: Low, Medium, and High.
- Triangular membership function was defined manually for each feature using their statistical ranges (min, mean, max). This converts crisp input values into fuzzy degrees of membership.

4.9 Rule Base Design

- A manual rule bank was constructed to relate combinations of input membership levels to the output surface roughness.
- With 3 membership levels for each of the 4 inputs, a total of 81 rules (3^4) were defined, assigning each rule to a linguistic output label.

4.10 Defuzzification

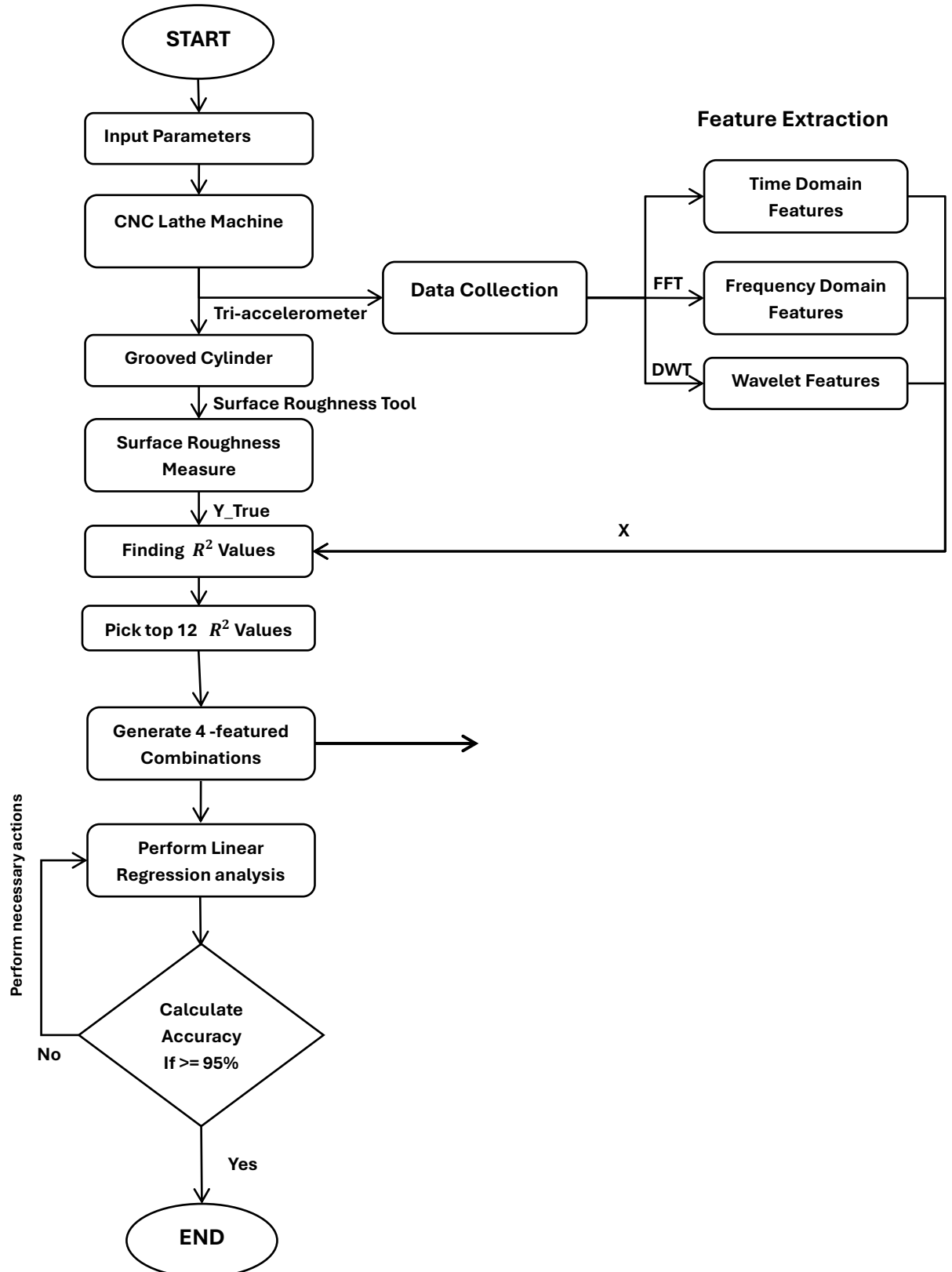
- The fuzzy output generated from the rule-firing strength is converted back into a crisp value using centroid-based defuzzification.
- This method calculates the center of gravity of the aggregated fuzzy set, resulting in a single predicted surface roughness value (Y_Pred) for each test part.

4.11 Prediction and Accuracy Evaluation

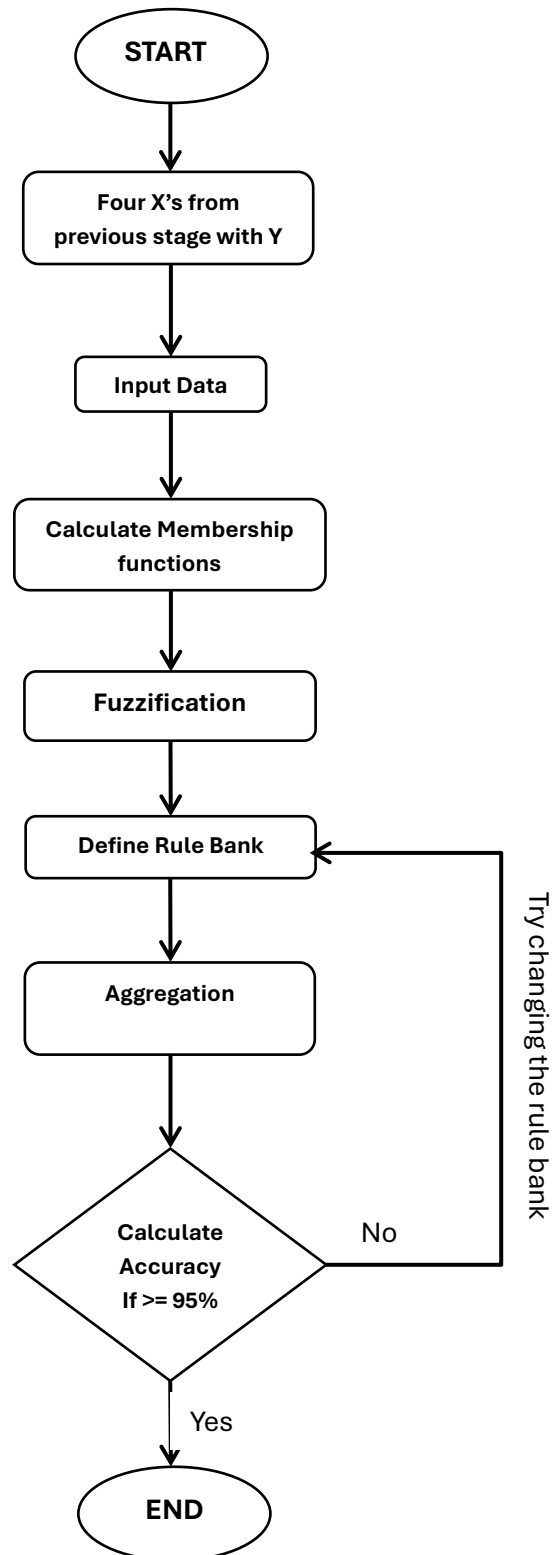
- The fuzzy logic model was executed for all test samples using the selected feature values.
- Predicted surface roughness values (Y_Pred) were compared with actual (Y_True), and accuracy was computed using a custom-defined formula.

5. Flow Diagram

Stage 1: Multi-Linear Regression



Stage 2: Fuzzy Logic Model



6. Results and Discussion

The below pictures are the outcomes of the predictive modeling approaches used in the project, including Multilinear Regression (MLR) and Fuzzy Logic, for estimating surface roughness (Y_True) based on vibration signals.

6.1 Stage 1 – Multilinear Regression Model Results

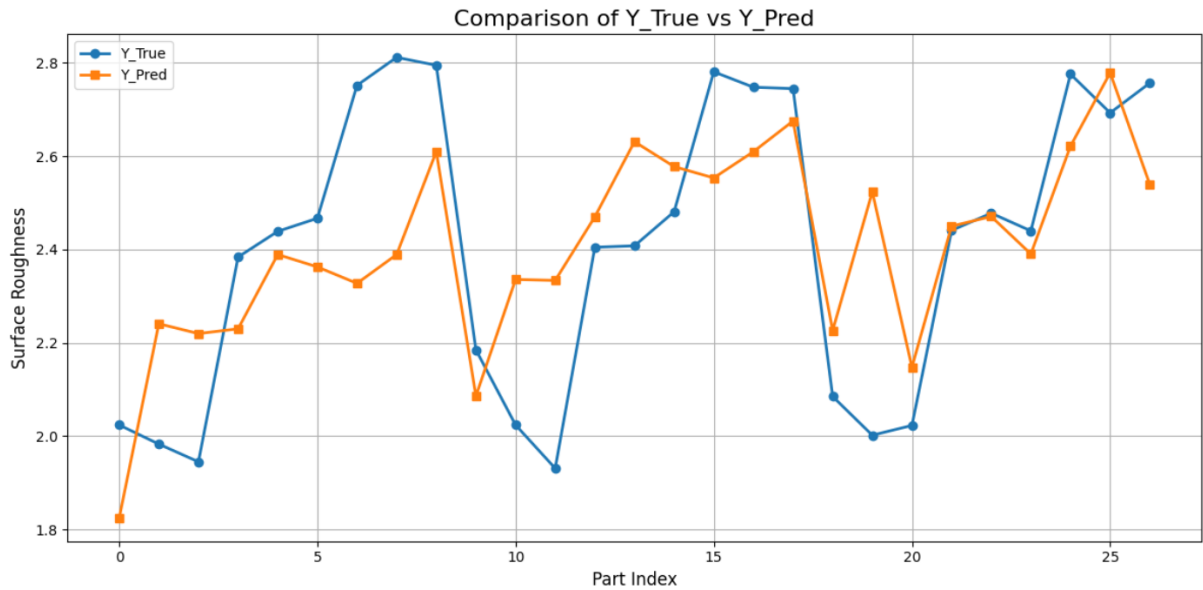
- The selected 4-feature combination was used to train a **Multilinear Regression (MLR)** model on the 27-part training dataset and predict surface roughness on the 16-part testing dataset.
- Predicted values (Y_Pred)** were computed for each test sample.
- The model showed a strong correlation between predicted and actual values.
- Model Accuracy:**
 - ✓ Training R²: **91.94%**
 - ✓ Testing Accuracy: **89.94%**

- Training Dataset Results:**

Channel 2 RMS	Channel 2 PEAK	Channel 2 KURTOSIS	Channel 2 PSD	Y_true	Y_Pred	Accuracy
0.011770926	0.052636	-0.002875375	52.11795345	2.024	1.824185	90.12773
0.01410921	0.061265	0.024684799	74.45467967	1.983	2.240928	86.99304
0.013974107	0.062621	0.041758737	73.29887407	1.945	2.219893	85.86667
0.014104	0.061758	0.032178289	74.71178829	2.384	2.230108	93.54482
0.015305678	0.064594	0.040822459	87.39656395	2.439	2.389345	97.96413
0.015348181	0.060402	-0.051669663	88.52835079	2.467	2.362768	95.77493
0.014777048	0.067429	0.0981051	81.94077208	2.752	2.327538	84.57622
0.015114588	0.067799	0.035722954	85.39847757	2.812	2.389307	84.96826
0.016367162	0.096151	0.109529871	101.374593	2.795	2.609946	93.3791
0.013195887	0.061635	0.113364022	65.5508018	2.185	2.085987	95.4685
0.014938777	0.062498	0.012450966	83.59832254	2.024	2.335834	84.59317
0.014587682	0.064224	-0.016084368	79.34431382	1.931	2.33386	79.13726
0.019978387	0.069401	0.037521401	150.1465578	2.405	2.470182	97.28973
0.020167474	0.077414	-0.264070465	151.8946964	2.408	2.631294	90.72699
0.01995147	0.076181	-0.099619803	149.0540016	2.481	2.577785	96.09897
0.0189329	0.074209	-0.016689949	134.8418478	2.781	2.553586	91.8226
0.018587453	0.069894	-0.161603526	128.7815556	2.748	2.609741	94.96873
0.018034173	0.085919	-0.108713219	122.0311761	2.745	2.675483	97.46751
0.014086122	0.060033	-0.015389127	74.56269709	2.085	2.226423	93.21714
0.016343945	0.072236	-0.024984633	99.86896828	2.002	2.523769	73.93764
0.013515917	0.055718	-0.084766238	68.50986601	2.023	2.147889	93.82655
0.015852945	0.063977	-0.132198788	94.30028383	2.441	2.450111	99.62673
0.015285635	0.07766	0.004047107	87.57972974	2.478	2.470852	99.71154
0.015409665	0.066319	0.075747435	88.873247	2.44	2.390582	97.97467
0.017086807	0.061882	-0.654956871	109.6879815	2.776	2.621722	94.44245
0.014814375	0.134364	0.152179665	82.30413846	2.693	2.778916	96.80964
0.016573895	0.069771	-0.099934393	102.6350933	2.757	2.539789	92.12149
						91.94208

- **Comparison For Y_True & Y_Pred:**

- This plot compares the actual values of surface roughness (Y_true) against the predicted values (Y_pred) for training 27 parts.



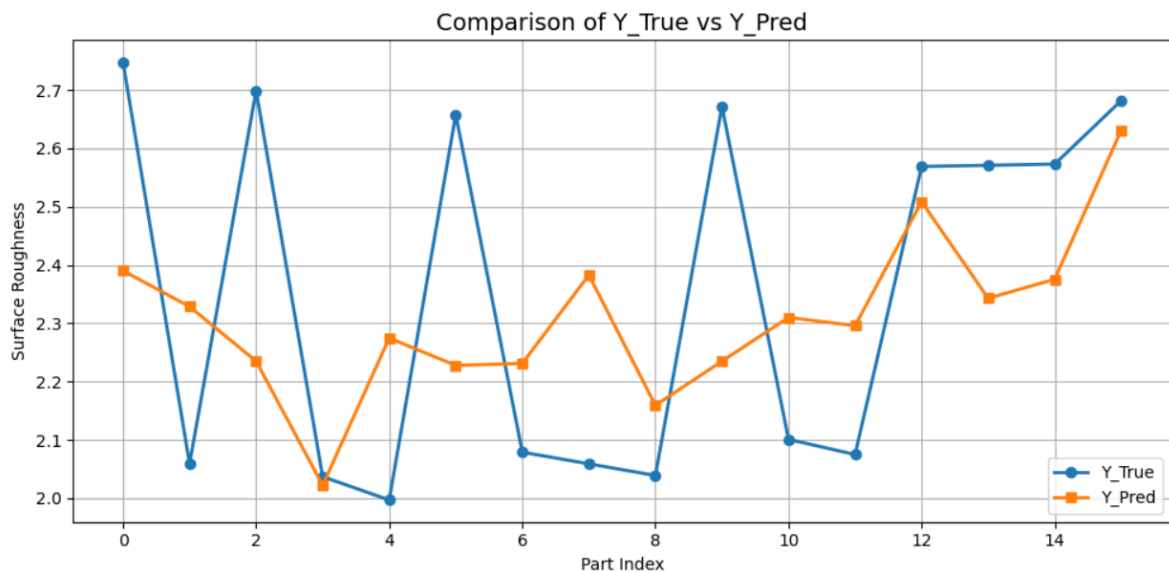
- The model is **predicting well** over all with 91.94% accuracy.

- **Testing Dataset Results:**

PART NO.	Channel 2 RMS	Channel 2 PEAK	Channel 2 KURTOSIS	Channel 2 PSD	Y_true	Y_pred	Accuracy
Part 1	0.015173649	0.070634	0.017856496	86.47221602	2.748	2.390176	86.97876
Part 2	0.014652638	0.071127	0.057769776	81.05832944	2.059	2.328678	86.90246
Part 3	0.014143332	0.062128	-0.001351196	74.98887603	2.699	2.235087	82.81167
Part 4	0.012847592	0.056088	-0.005292631	61.84060361	2.037	2.022213	99.27408
Part 5	0.014256159	0.064594	0.046837532	75.91364543	1.997	2.274729	86.09269
Part 6	0.014206703	0.05806	-0.003349759	75.33725834	2.658	2.227722	83.81196
Part 7	0.014464669	0.056828	-0.036421296	78.55824249	2.079	2.231318	92.67348
Part 8	0.015660135	0.06447	-0.05440355	92.24655985	2.059	2.382839	84.27202
Part 9	0.01372812	0.059416	-0.036692079	70.83927577	2.039	2.159343	94.09794
Part 10	0.014080459	0.064717	0.027398895	74.58378611	2.672	2.235059	83.64743
Part 11	0.014785421	0.063238	0.023019453	81.98409902	2.101	2.310083	90.04843
Part 12	0.014561979	0.063607	-0.022587904	79.36754877	2.075	2.295832	89.35748
Part 13	0.017491753	0.069155	-0.088133592	114.8156116	2.569	2.508508	97.6453
Part 14	0.014958689	0.067799	0.191689618	84.28530792	2.571	2.342952	91.12999
Part 15	0.015411109	0.064347	-0.06859144	89.00716264	2.573	2.375578	92.32718
Part 16	0.019427342	0.089248	-0.360499136	141.5941746	2.682	2.631337	98.111
							89.94887

- **Comparison For Y_True & Y_Pred:**

- This plot compares the actual surface roughness values (Y_True) against the predicted values (Y_Pred) for testing 16 parts. It helps evaluate how well your prediction model is performing.



- The average accuracy was **89.95%**, which is quite good for predictive tasks.

6.2 Stage 2 – Fuzzy Logic Model Results

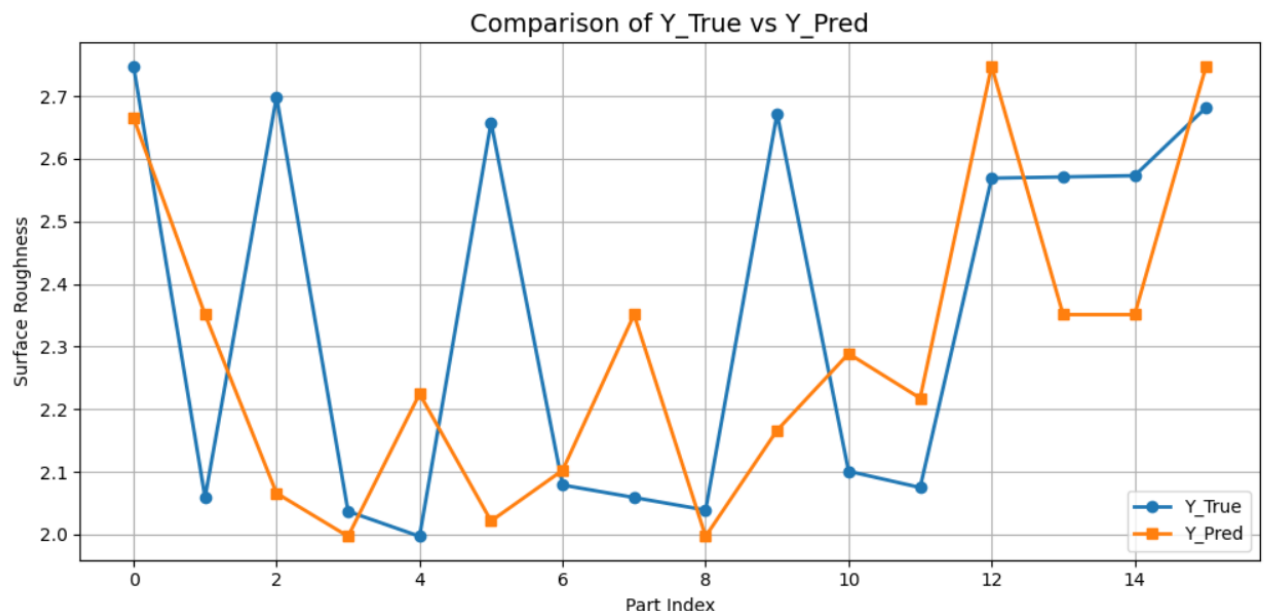
- To improve handling of non-linear and uncertain relationships, a **fuzzy logic model** was developed using the same 4 selected features.
 - **Fuzzification:** Each input was mapped to Low, Medium, and High membership levels.
 - **Rule Base:** A total of 81 fuzzy rules (3^3 for 4 inputs) were designed manually.
 - **Defuzzification:** Centroid method was used to convert fuzzy outputs into crisp Y_Pred values.
 - **Accuracy:**
 - ✓ Testing Accuracy: **90.20%**

- **Testing Dataset Results:**

Channel 2 RMS	Channel 2 PEAK	Channel 2 KURTOSIS	Channel 2 PSD	Y_true	Y_pred	Accuracy (%)
0.015173649	0.070634	0.017856496	86.47221602	2.748	2.6652	96.99
0.014652638	0.071127	0.057769776	81.05832944	2.059	2.3511	85.81
0.014143332	0.062128	-0.001351196	74.98887603	2.699	2.0659	76.54
0.012847592	0.056088	-0.005292631	61.84060361	2.037	1.997	98.04
0.014256159	0.064594	0.046837532	75.91364543	1.997	2.2246	88.6
0.014206703	0.05806	-0.003349759	75.33725834	2.658	2.0215	76.05
0.014464669	0.056828	-0.036421296	78.55824249	2.079	2.1025	98.87
0.015660135	0.06447	-0.05440355	92.24655985	2.059	2.3511	85.81
0.01372812	0.059416	-0.036692079	70.83927577	2.039	1.997	97.94
0.014080459	0.064717	0.027398895	74.58378611	2.672	2.1664	81.08
0.014785421	0.063238	0.023019453	81.98409902	2.101	2.2888	91.06
0.014561979	0.063607	-0.022587904	79.36754877	2.075	2.2177	93.12
0.017491753	0.069155	-0.088133592	114.8156116	2.569	2.748	93.03
0.014958689	0.067799	0.191689618	84.28530792	2.571	2.3511	91.45
0.015411109	0.064347	-0.06859144	89.00716264	2.573	2.3511	91.38
0.019427342	0.089248	-0.360499136	141.5941746	2.682	2.748	97.54
						90.21

- **Comparison For Y_True & Y_Pred:**

- ✓ The below plot shows the comparison between actual values and predicted values for the fuzzy logic model in which the blue line indicates actual and orange line indicates predicted values.



- ✓ The model predicted the outcomes with **90.20%** accuracy, which is quite good.